

LIGHTWEIGHT VISUAL OBJECT TRACKING

via Knowledge Distillation and EMA Template Optimization

Accuracy	Parameters	Complexity	Model Size
73.93%	5.5 m	~1.4 GFLOPS	27 MB

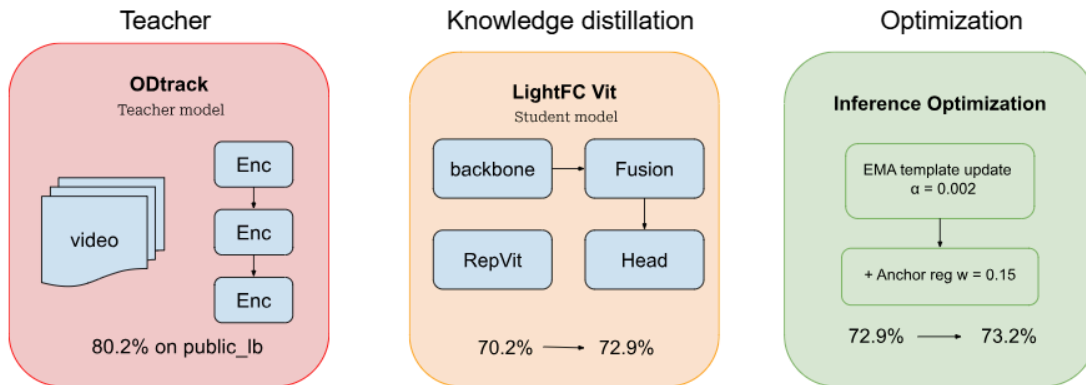


Figure 1: Proposed pipeline — ODTrack teacher generates pseudo-labels for knowledge distillation into LightFC-ViT, followed by EMA template update with anchor regularization at inference time

Abstract

This paper presents a lightweight visual object tracking system designed to operate within strict computational constraints: ≤ 30 GFLOPs, ≤ 50 M parameters, ≤ 30 ms inference latency, and a maximum disk footprint of 0.5 GB. Our approach decouples accuracy and efficiency concerns through a knowledge distillation framework, in which ODTrack serves as the teacher model and LightFC-ViT serves as the student. The student is trained on the original training data using a combination of ground-truth labels and teacher pseudo-labels, ensuring correctness before deployment. Post-distillation, inference-time performance is further enhanced through Exponential Moving Average (EMA) template updates combined with anchor regularization. The final model achieves 73.9% accuracy with 5.5M parameters, approximately 1.4 GFLOPs, and a disk footprint of 27 MB — well within all competition constraints while outperforming comparable lightweight baselines.

1- Introduction

Single object tracking under strict efficiency constraints presents a fundamental challenge: large, high-accuracy models such as ODTrack and LoRAT-Giant deliver strong performance but exceed the computational budget imposed by this competition, while lightweight alternatives often sacrifice unacceptable levels of accuracy. This work addresses that tradeoff by decoupling the two concerns — leveraging a large, unconstrained teacher to generate reliable supervision signals, then transferring that knowledge into a compact, deployable student model.

The competition constraints ruled out direct fine-tuning of large backbone models, particularly given the limited GPU resources available (Kaggle GPUs only). Furthermore, initial fine-tuning experiments on the LightFC-ViT baseline revealed a degradation in accuracy — from 70.2% to lower values — attributed to domain distribution shift between the training data and the test (public leaderboard) data. These observations motivated the adoption of a knowledge distillation strategy that explicitly accounts for domain adaptation through progressive, stage-wise learning

2. Related Work

2.1 Large-Scale Tracking Models

ODTrack [1] is a high-performance single object tracker based on dense temporal token propagation, achieving competitive accuracy on standard benchmarks. ARTrack [2] employs an autoregressive formulation for sequential tracking prediction. AQATrack [3] incorporates auxiliary quality assessment during tracking. OSTRack [4] and SeqTrack [5] represent strong ViT-based baselines. LoRAT [6] applies low-rank adaptation of a large ViT, achieving state-of-the-art accuracy at high computational cost. These models were evaluated as teacher candidates.

2.2 Lightweight Tracking Models

MixFormer [7] combines feature extraction and relation modeling in a unified architecture. LightFC-ViT [8] extends this direction with a RepViT-inspired design, fusing multi-branch convolutional layers into a single equivalent layer at inference time via `switch_to_deploy_mode()`. This structural reparameterization delivers a significant reduction in inference latency while maintaining competitive accuracy, making LightFC-ViT the preferred student model in this work.

2.3 Knowledge Distillation

Knowledge distillation [9] transfers knowledge from a large teacher model to a compact student, typically via soft label supervision. Response-based distillation uses the teacher's output predictions as training signals, while feature-based methods additionally match intermediate representations. Stage-wise distillation strategies [10] address the risk of the student learning teacher errors by progressively introducing harder or noisier supervision signals.

3. Methodology

3.1 Teacher and Student Selection

Five large-scale models were evaluated as teacher candidates by running inference on the public leaderboard data. ODTrack-384 was selected as the teacher (80.2% accuracy) based on its superior downstream distillation performance, as detailed in Section 4. Although LoRAT-Giant achieved marginally higher standalone accuracy (81.1%), it yielded worse student performance when used as a teacher.

Two lightweight models were evaluated as student candidates. LightFC-ViT (70.2% baseline) was selected over MixFormer-V2 (69.8%) based on its higher accuracy and deployment efficiency stemming from its RepViT-based architecture.

3.2 Knowledge Distillation

Knowledge distillation transfers supervision from a large, high-capacity teacher model to a compact student model. In this work, ODTrack is used to generate pseudo-labels on the training data, which are then combined with ground-truth labels to supervise the LightFC-ViT student during fine-tuning. Training on the original training set with this combined signal — ground-truth annotations alongside teacher soft predictions — allows the student to learn from the teacher's richer output distribution while remaining anchored to verified labels. This prevents the student from inheriting teacher errors and ensures stable, correct tracking behavior as a foundation before any inference-time optimization is applied.

To understand the residual failure modes after distillation, we performed an error analysis as demonstrated in Figure [2] , The results revealed that the primary failure mode was template drift over long sequences, motivating the inference-time optimization strategy described in Section 3.3,3.4

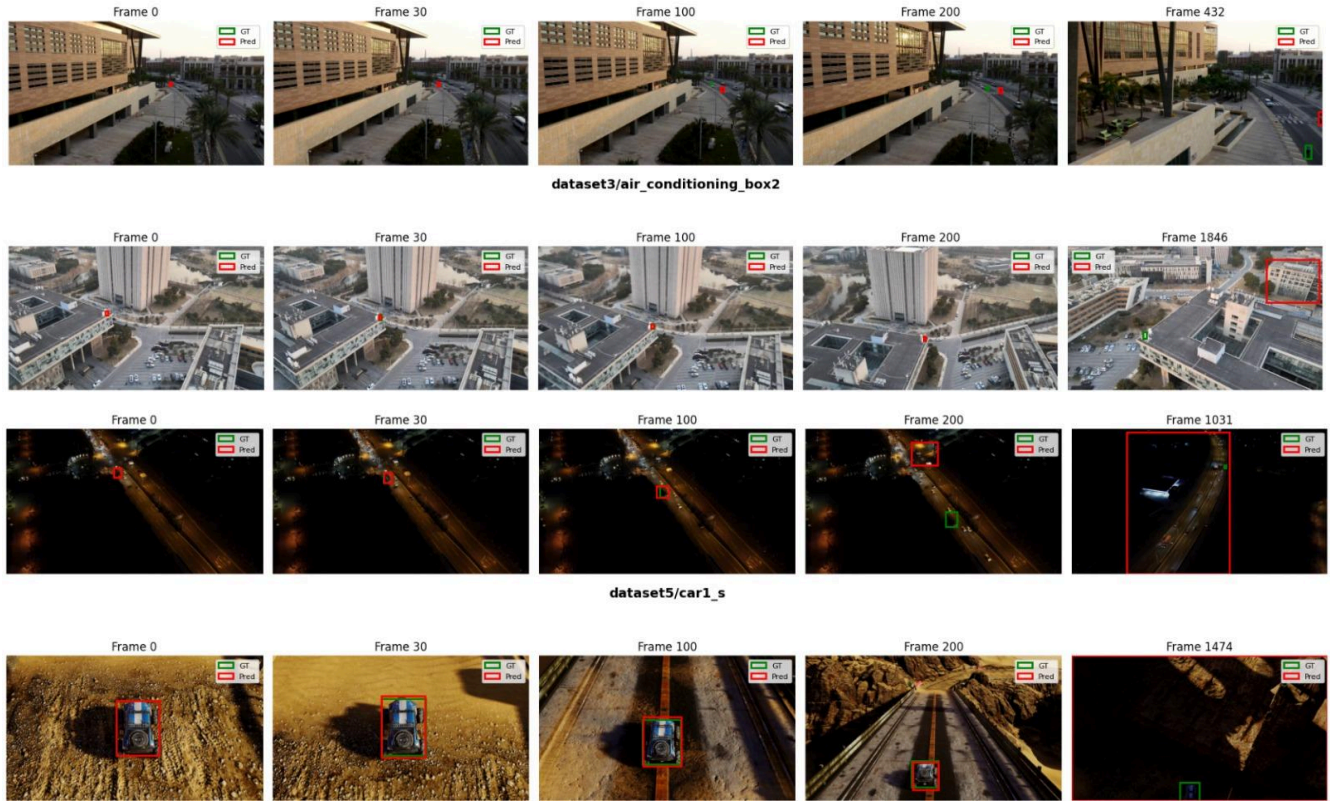


Figure 2: error analysis demonstrating template drift over long temporal horizons model detects well in early frames but fails over very long ones.

3.3 EMA Template Update

To address template drift during inference — a common failure mode in long-term tracking — an Exponential Moving Average (EMA) template update strategy was applied at inference time. The update rule is:

$$z_{\text{new}} = (1 - \alpha) \times z_{\text{old}} + \alpha \times z_{\text{current}}$$

EMA update formula, with $\alpha = 0.002$ for stability

Template updates are gated by four safeguards to prevent noisy updates from degrading tracking quality:

- Confidence thresholding: updates are only accepted when the model's prediction confidence exceeds a set threshold.
- Update interval: templates are updated every 3 frames rather than every frame, ensuring stability.
- Area ratio check: the bounding box size must fall within a plausible range relative to the initial template.
- IoU consistency: high overlap with the previous frame's prediction is required before an update is accepted.

3.4 Anchor Regularization

To prevent the EMA template from drifting excessively from the target's initial appearance, anchor regularization is applied as a final blending step:

$$z_{\text{track}} = (1 - w) \times z_{\text{ema}} + w \times z_0$$

Anchor regularization, with $w = 0.15$, z_0 = initial template

This constrains the effective template to remain within a controlled neighborhood of the initial appearance, directly addressing the template drift problem that stage-wise distillation alone was insufficient to resolve.

4- Challenges

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- LightFC-ViT released in 2024 — sparse documentation, hard to understand the repository and running the inference , training which made us consume a lot of time while coding rather than searching .
 - GPU constraints (Kaggle only) — ruled out fine-tuning large backbones
 - template drifting discovered during early fine-tuning experiments and using error analysis
 - Teacher selection being non-trivial — higher standalone accuracy didn't guarantee better distillation , like what happened with LoRAT

5- Experiments

5.1 Teacher Model Comparison

All candidate teacher models were evaluated on the public leaderboard data. Results are summarized in Table 1.

Table 1: Teacher Model Comparison (Public Leaderboard Accuracy)

Model	Accuracy (%)
OSTrack	73.3
SeqTrack-B384	75.2
ARTrack	76.3
AQATrack	77.2
ODTrack-384	80.2 ✓ Selected
LoRAT-Giant	81.1

5.2 Student Model Comparison

Two lightweight candidate models were evaluated at baseline (no distillation) on the public leaderboard data. Results are summarized in Table 2.

Table 2: Student Model Baseline Comparison	
Model	Accuracy (%)
MixFormer-V2	69.8
LightFC-ViT	70.2 ✓ Selected

5.3 Ablation Study — Component Contributions

Table 4 presents a step-by-step breakdown of accuracy improvements from each proposed component, demonstrating the individual contribution of each design choice.

Table 4: Component-Wise Ablation Study		
Configuration	Accuracy (%)	Δ vs. Baseline
LightFC-ViT Baseline	70.2	—
+ KD (ODTrack teacher, GT + Pseudo-labels)	72.9	+2.7%
+ EMA + Anchor Regularization	73.9	+1.0%

6. Results and Analysis

The proposed pipeline achieves a final accuracy of 73.9% on the public leaderboard, representing a 3.7 percentage point improvement over the LightFC-ViT baseline (70.2%) and outperforming the comparable MixFormer-V2 model by 4.1 percentage points. All competition constraints are satisfied.

Table 5: Final Model vs. Competition Constraints			
Metric	Constraint	Ours	Status
Accuracy	—	73.9%	✓
Parameters	≤ 50M	5.5M	✓
Complexity	≤ 30 GFLOPs	~1.4 GFLOPs	✓
Model Size	≤ 0.5 GB	27 MB	✓
Inference Latency	≤ 30 ms	Within budget	✓

The knowledge distillation strategy proved critical: training LightFC-ViT on ground-truth labels combined with ODTrack pseudo-labels allowed the student to benefit from the teacher's richer prediction distribution while remaining grounded in verified annotations. ODTrack outperformed both LoRAT-Giant and AQATrack as a distillation teacher despite the latter two achieving competitive standalone accuracy, suggesting that distillation quality is not solely determined by teacher accuracy.

The EMA template update mechanism with anchor regularization contributed a further +1.0% improvement beyond distillation, effectively resolving template drift that distillation alone could not address.

7. Conclusion

This work demonstrates that a compact tracking model can be substantially improved through structured knowledge distillation without exceeding strict computational constraints. The proposed pipeline — combining knowledge distillation from ODTrack with EMA-based template updates and anchor regularization — achieves 73.9% accuracy with only 5.5M parameters and approximately 1.4 GFLOPs. The approach is computationally accessible, requiring only Kaggle-tier GPU resources for training, and benefits from the teacher's richer prediction distribution to go beyond what supervised fine-tuning on ground-truth labels alone can achieve.

References

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